



Smartphone problems

Here is a list of 12 most commonly faced problems with smartphones which you are likely to face, if you haven't already. <http://dgit.in/SmartProbs>

Google blocker

Google Now has a new feature to block out content that's irrelevant to you with a simple flick. Here's how you do it. <http://dgit.in/GBlocker>

ally, games began to be designed with this approach in mind, leading to Epic Games' Unreal Engine and other such game engines. A Game Engine behaves as a higher level library that implements abstract concepts like entities, material, camera, lights using the capabilities of the lower Graphics API, like mesh, texture, shaders etc.

At present, both Games and Game Engines are incredibly complex and expensive to make. Most Game development studios have stopped focussing on developing game engine technology and prefer to use existing options in the market such as Unity3D and Unreal Engine. Consequently, graphics drivers have also become substantially complicated. As a gamer, it might escape one's attention but one look at the size of the installer tells us the amount of

graphics compared to the open standard that PHIGS was.

But due to competition from Sun Microsystems, IBM and Hewlett-Packard's own 3D hardware being brought into the market with extensions made using the PHIGS open standard, SGI began to lose market share. In an effort to regain the same, they turned their own standard open source and renamed it to OpenGL.

OpenGL was more than a mere rebranding. It overcame drawbacks in IRIS GL that prevented an application from implementing and using a feature that was intrinsically unsupported by the underlying hardware. The control of OpenGL was handed over to OpenGL Architecture Review Board, which was an industry consortium of companies like Apple, IBM, Intel etc.

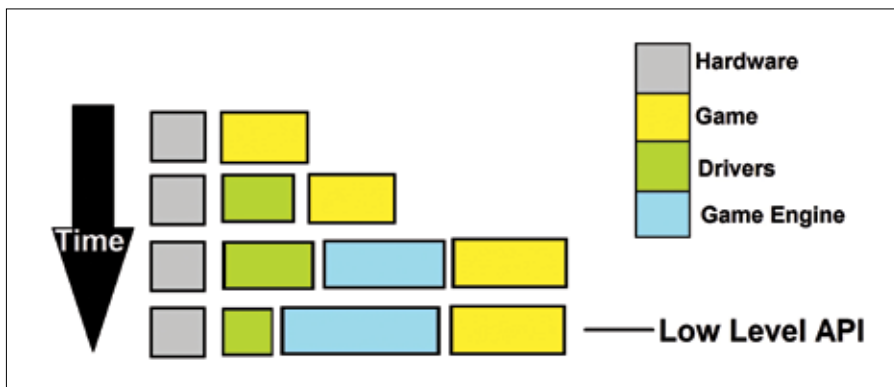
DirectX. Incidentally, DirectX did support OpenGL. For example, if a developer chose to use the OpenGL 3D graphics API, other DirectX API's were combined with it to be used for games because there were some features of DirectX that OpenGL did not provide, like joystick support.

For the console side of Microsoft's ventures, DirectX formed the basis for the entire Xbox range's API. The entire API was developed by Microsoft in conjunction with Nvidia which built the hardware for the Xbox. In fact, the Xbox was originally named DirectXbox, but later renamed for commercial purposes.

On one side, as Microsoft has been gunning their DirectX efforts, the control of the OpenGL was passed to another industry consortium called the Khronos group in July 2006. The Khronos group consists of a much larger scale of companies like AMD, Apple, Google, Mozilla, IBM, Samsung, RedHat, Microsoft, Oculus, Panasonic, Nintendo, Pixar, Lucasfilm, Amazon, Valve, Unity, Nokia, Nvidia (We are sure you get the drift!). Further development of OpenGL was handled by the Khronos Group.

Metal for iPhones and other Apple devices debuted in iOS8 and aimed to bring support and benefits of similar APIs in other platforms to the Apple Ecosystem. Metal's features differentiated itself from OpenGL by mainly reducing the CPU overhead and explicit synchronization between CPU and GPU, which basically reduced the number of tasks to be done by the CPU leaving it free to take up additional load.

AMD's proprietary solution, the Mantle API was introduced to the market when the only other major API options were Microsoft's Direct X on Windows and OpenGL on many other platforms. The low level API provided significant benefits such as a highly reduced CPU overhead, Native Multi-GPU support, Explicit command buffer control etc. To the player, the reduced overhead and fewer direct calls to the CPU meant that the players with high-end GPU's would be getting much better frame rates than before as the GPU was no longer held back by the CPU. But, quite similar to Metal, Mantle was also restricted to work with AMD's own hardware, for reasons attributed to lack of industry support or internal decisions,



The above diagram shows how the interaction between the hardware and the application has evolved to the present

code that goes into something that does not directly contain any of the actual graphics or media assets. That is because each driver has to implement a lot of APIs (DirectX versions, OpenGL versions etc) and also has to be backward compatible. This complexity and enforced compatibility often leads to unnecessary overhead and unwanted bugs.

The story of the predecessors

The history of Graphics APIs began much before the 90s but one of the main boosts was received during that era. PHIGS, the standard being used by the industry for Graphics development, was gradually replaced by the IRIS GL API by Silicon Graphics, mainly due to ease of use and immediate mode rendering, which was more suitable for rendering more complex

In the mid-90s, Microsoft had released the first version of its own proprietary API package as Windows Games SDK which they later named DirectX. Even though initial adoption of DirectX was slow with programmers, publicity efforts along with thorough testing and compatibility measures from Microsoft gradually grew its popularity within the Windows Platform. Prior to DirectX, Microsoft had included OpenGL support for WindowsNT targeting engineering and CAD applications and the Direct3D API in DirectX was originally intended to be a low level support for usage like gaming. Over time, as graphics grew more complicated, OpenGL also included multimedia support and programmers got the option to choose between the open-source OpenGL and Microsoft-only